

Microstructure and electrochemical properties of LBO-coated Li-excess $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material at elevated temperature for Li-ion battery

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Received 14 July 2005; received in revised form 6 October 2005; accepted 13 October 2005

Available online 28 November 2005

Abstract

An amorphous glass film, $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ (LBO) glass, was coated on the surface of the cathode material by solution method. The Li-excess cathode powder $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ derived from co-precipitation method was calcined with various weight percentage of the surface modified lithium boron glass. Fine powders with distinct particle size, size distribution and morphology were fabricated. The electron probe microanalyzer (EPMA) was employed to evaluate the composition of LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$. The morphology was observed with a field emission scanning electron microscope (FE-SEM), and the particle size in the range of several microns measured by laser scattering. The electrochemical behavior of the cathode powder was examined by using two-electrode test cells consisted of a cathode, metallic lithium as anode, and an electrolyte of 1 M lithium hexafluorophosphate (LiPF_6). Cyclic charge/discharge testing of the coin cells, fabricated by both LBO-coated and base $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ material were conducted. The LBO-coated cathode powder with the fading rate of only 7% after 25 cycles showed better cycleability than the base one with the fading rate of 17% after 25 cycles, particularly at higher temperature. It is demonstrated that the employment of LBO glass coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material exhibited higher discharge capacity and significantly reduced the fading rate after cyclic test.

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Keywords: Li-ion battery; LiMn_2O_4 cathode material; $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass; Surface modification; High temperature performance

1. Introduction

There are various types of cathode materials in Li-ion rechargeable batteries with low cost, high stability and good electrochemical performance, such as high discharge capacity and low fading rate at high temperature. LiCoO_2 [1] with layer (R3m) structure has been widely used in the electronic portable devices, such as cellular phone, notebook, and personal digital assistance (PDA) in recently years. Other cathode materials, like LiNiO_2 and LiNiCoO_2 [2,3] with the layer (R3m) structure, LiFePO_4 and LiMnPO_4 [4,5] with olivine (Pmnb) structure, LiNiVO_4 and LiCoVO_4 [6] with the inverse spinel (Fd3m) structure and LiMn_2O_4 [7–10] with spinel (Fd3m) structure were also investigated extensively. Among cathode materials, LiMn_2O_4 with stable spinel structure in Li-ion battery has become the promising cathode material for commercial usage owing to the

lower cost than LiCoO_2 and better thermal stability than LiNiO_2 . In addition, Mn is non-toxic, environmentally friendly and relatively abundant.

Before commercial usage, however, LiMn_2O_4 cathode material might have poor cycling behavior, particularly at high temperature. First, the Jahn-Teller distortion [11] caused by Mn^{3+} ion would lead to the cycleability degradation in LiMn_2O_4 during cycling. It was reported that the doping of LiMn_2O_4 with Li and transition metal, such as Cr, could successfully improve the cycleability [12]. The formation of other phases, such as Mn_2O_3 , during conventional high temperature calcination could be suppressed by the excess lithium and chromium doping. However, Mn dissolution resulted from some side reactions occurred at the interface between the electrode and the electrolyte during the charge/discharge process [13]. As the cathode electrode contacted with the Li-based electrolyte directly in Li-ion batteries, Mn dissolution was induced by the generation of acids like HF, which was resulted from the reactions of fluorinated anions with the manufactures of instable Li-based salt [14,15] and solvent oxidation [16,17]. Particularly, at higher temperature as 55 °C,

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the Mn dissolution and side reactions leading to the poor electrochemical performance could become more and more severe.

In order to solve this problem, surface modification of the cathode electrode is an effective way to reduce the side reactions. The surface treatment could decrease the surface area to retard the side reactions between the electrode and electrolyte, and to further diminish the Mn dissolution during cycling test. In literature, the utilization of metal oxides [18–20] and organic/inorganic materials [21–23] for surface modification in both cathode and anode electrode could improve the electrochemical property of Li-ion battery. In Amatucci's study [24], lithium boron oxide was proposed as a proper surface treatment material for cathode material. In fact, lithium boron oxide has been widely studied as a solid lithium ionic conductor with good lithium ionic conductivity [25,26]. In addition, the molten lithium boron oxide with good wetting property and relatively low viscosity provide easy manufacturing even with slight amount of coating material [27].

In this study, an effective approach was employed to decrease the Mn dissolution and side reactions occurred at the interface between the cathode electrode and electrolyte. It was aimed to obtain the satisfactory electrochemical property for Li-ion rechargeable batteries. $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass was chosen as the coating material, which was coated on the surface of the cathode material by solution method. In addition, the correlation between microstructure and electrochemical property of LBO glass coated LiMn_2O_4 cathode material was examined and discussed.

2. Experimental procedure

The precursor of $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass was synthesized by mixing LiOH and H_3BO_3 and stirred gently with the molar ratio of 1:2 in methanol. $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode powder synthesized by co-precipitation method [12] was added into the solution and heated at 50–60 °C until the solvent was evaporated completely. The mixture powders of the precursor of the LBO glass and $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ were then calcined at 500 °C for 10 h and the calcined temperature was determined by a preliminary thermal analysis [28]. The weight percent of the LBO glass was 0–1 wt.% before heat-treatment, and evaluated to be 0–0.5 wt.% after calcination. Table 1 lists the sample designation of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode powders under various conditions.

The compositions of the LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode material were analyzed with an electron probe microanalyzer (EPMA, JXA-8800M, JEOL, Japan). However, since lithium

could not be detected in the quantitative analysis of EPMA, the contents of boron, manganese and oxygen were evaluated first through ZAF technique [29] in EPMA and then the amount of lithium was obtained by difference and normalization approach [30]. The X-ray diffractometer (XRD, Rigaku, D/MAX-B, Japan) with a wavelength of Cu $K\alpha$ ($\lambda = 1.5406\text{\AA}$) was used to identify the crystal structure and phases of the LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode material operated at 30 kV and 20 mA from 15° to 70°. The particle morphology, as well as particle size and distribution of the LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ powder were examined using a field-emission scanning electron microscope (FESEM, JSM-6500F, JEOL, Japan) at an accelerating voltage of 15 kV and the laser scattering (Horiba, LA 300, Japan), respectively. Transmission electron microscope (TEM, JEM-2010, JEOL, Japan) was also utilized to evaluate the LBO film thickness coated on the surface of the cathode powders as well as the phase confirmation for both the coating layer and cathode powder.

The electrochemical behavior of the LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode electrode was measured by two-electrode test cells consisted of the LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ as cathode, metallic lithium anode and an electrolyte of 1 M LiPF_6 in a 1:1 (volume ratio) mixture of ethylene carbonate/dimethyl carbonate (EC/DMC). The cell fabrication was carried out in a specially designed argon-filled chamber with low oxygen pressure ($\text{O}_2 < 2$ ppm) and low moisture ($\text{H}_2\text{O} < 2$ ppm). The constructed 2016 coin cells was then cycled at 0.1 C rate for the first cycle and at 0.2 C rate from the second cycle. The electrochemical impedance spectroscopy (EIS, EIS300, Gamry, USA) was used to measure the AC impedance of the 2016 coin cell before and after cycling test. Furthermore, the 2016 coin cell after cycling was disassembled and the cathode electrode was analyzed for the variation of the structure of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode material after cycling.

About 10 mg each of base and LBO-coated lithium manganese oxide powder were immersed in 10 ml 1 M LiPF_6 with a 1:1 (volume ratio) mixture of EC/DMC for 3 days. About 0.1 M HCl was then used to extract the dissolved Mn^{2+} in the electrolyte into the water phase [31]. The amounts of Mn in diluted HCl were quantitatively confirmed by inductively coupled plasma (ICP).

3. Results and discussion

3.1. Microstructure characterization

Fig. 1 shows FE-SEM microphotographs of LBO-coated and uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$. In Fig. 1(a), the facets of the uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ with originally sharp edges appear rather smooth. However, with a more detailed view as shown in Fig. 1(b) and (c), the facet of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ with sharp edges was obviously covered by a rough coating layer. It is argued that the rough coating layer on the surface of cathode powder was LBO glass. To confirm the composition of the coating layer, an energy dispersive spectrometers (EDS) attached on the FE-SEM was used to identify the elements in the coating film. From Fig. 1(d), manganese, oxygen, carbon and boron were found in

Table 1
Sample designations of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode powders under various conditions

	Before cycling	25 cycles at 25 °C	25 cycles at 60 °C
Base $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$	A1	A2	A3
0.1 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$	B1	B2	B3
0.3 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$	C1	C2	C3
0.5 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$	D1	D2	D3

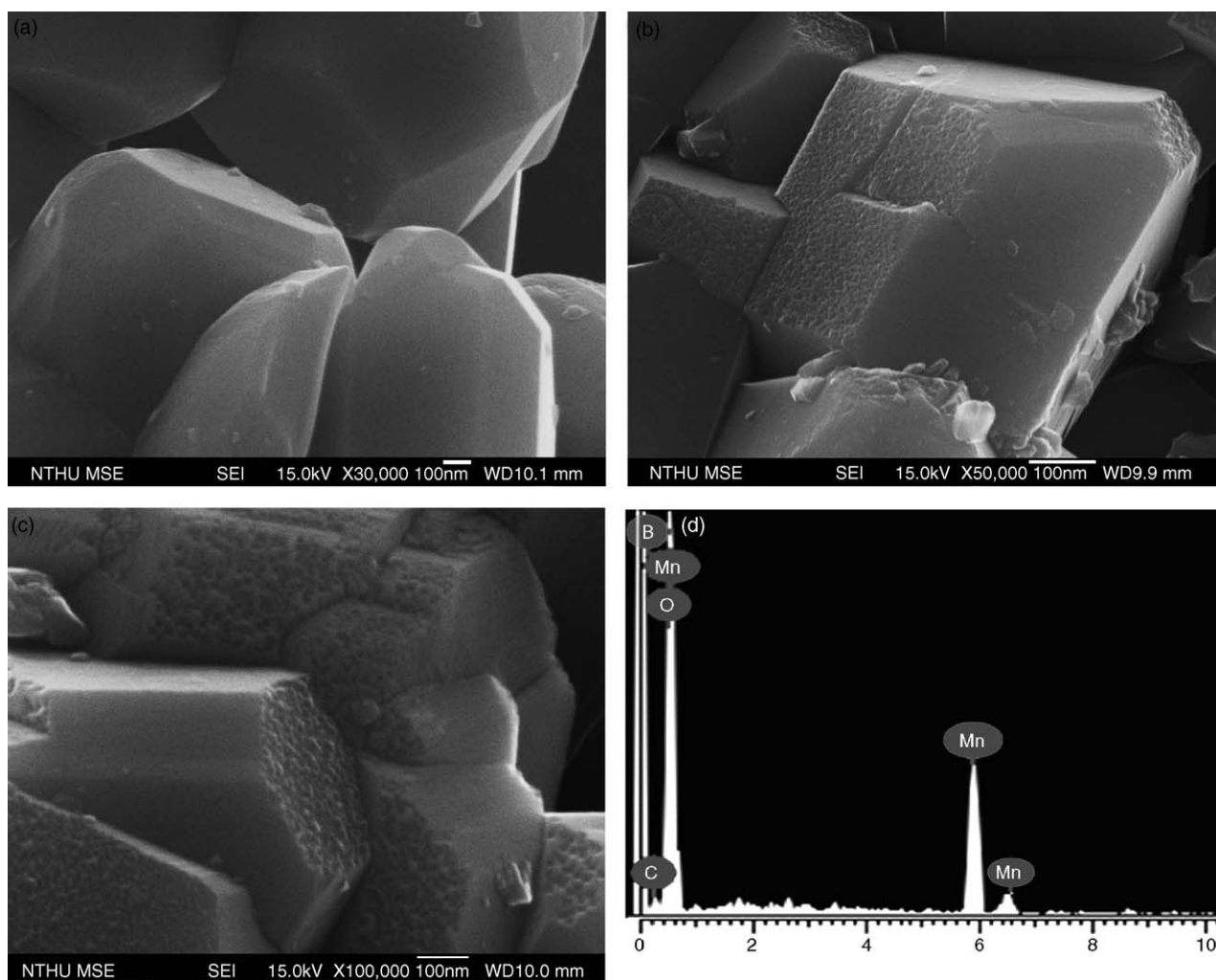


Fig. 1. FESEM images of various powders calcined at 500 °C for 10 h: (a) base $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, (b) 0.3 wt.%, (c) 0.5 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, and (d) EDS spectra of the amorphous coating layer.

EDS spectra, while carbon spectrum was contributed from the carbon paint in the sample preparation. On the contrary, only manganese, oxygen and carbon were detected in the EDS of the uncoated LiMn_2O_4 .

From TEM image in Fig. 2, it was observed that an amorphous film with the thickness in the range of 60–100 nm was coated on the surface of $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode powder. For a comparison, a preliminary study of LBO-coated powder was fabricated by the solid state method [32]. It appeared that the amorphous layer coated on the surface of $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode powder by solution method was more homogeneous than the solid-state one.

A recent study [33] revealed that the X-ray spectra for various amount of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ were almost identical to that for the uncoated one. Spinel structure was demonstrated without any other phases. Based on the results of XRD, the lattice constant exhibited small deviation, indicating that the structure of spinel cathode material was maintained after surface modification. This suggests that LBO glass was coated only on the surface rather than penetrating into the spinel structure. It was believed that the phase of LBO glass coated on the surface of cathode material was amorphous.

On the basis of the FESEM micrographs, XRD spectra and TEM analysis, it was demonstrated that the amorphous LBO glass could be coated homogeneously on the surface of $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

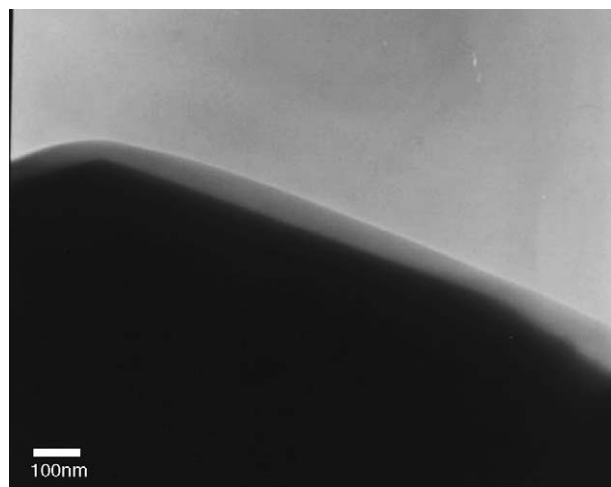


Fig. 2. TEM image of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ calcined at 500 °C for 10 h.

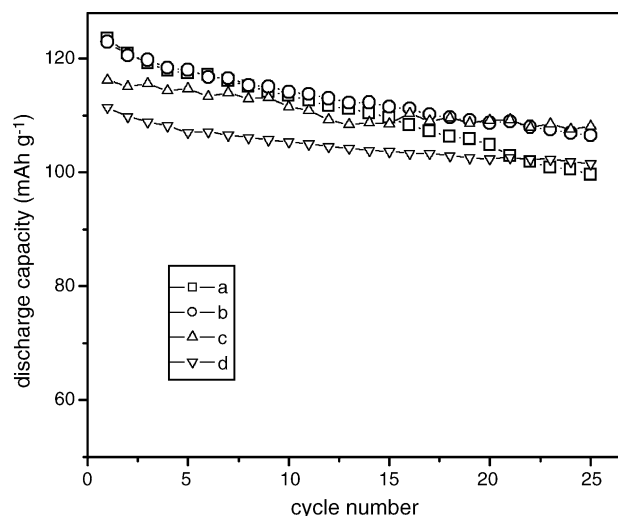


Fig. 3. Specific discharge capacity of various powders calcined at 500 °C for 10 h followed by 25 cycles at 25 °C: (a) uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, (b) 0.1 wt.%, (c) 0.3 wt.%, and (d) 0.5 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

3.2. Electrochemical property

The 2016 coin cell was measured galvanostatically between 3 and 4.2 V with the assembly of $\text{Li}/\text{EC} + \text{DMC} + \text{LiPF}_6/\text{LiMn}_2\text{O}_4$ at 0.1 C rate for the first cycle and then at 0.2 C rate after the second cycle. The fading rate was calculated from the test of the base $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ and various amount of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ calcined at 500 °C for 10 h, as shown in Fig. 3.

The initial discharge capacity of sample A1, i.e. uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ was measured as 123.5 mAh g^{-1} and decayed to 103.7 mAh g^{-1} after 25 cycles, as shown in Fig. 3(a), which exhibited a fading of 16%. Fig. 3(b)–(d) show that the initial discharge capacity for the samples B2, C2 and D2 was 122.5, 116.2 and 111.4 mAh g^{-1} , respectively, at the first cycle, and remained 87%, 93% and 92%, respectively, of its original value after 25 cycles. In other words, the fading percentage per cycle was 0.65%, 0.52%, 0.28% and 0.32% for samples A1, B2, C2 and D2, respectively.

In Fig. 3, the fading rate of the $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode material was obviously reduced with the increase of LBO glass content. Among all LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, 0.3 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode material exhibited the best electrochemical property.

In order to verify the effect of LBO coating layer upon decreasing the dissolution of spinel LiMn_2O_4 cathode, the Mn dissolution experiment was also carried out in this study. ICP results for Mn dissolution of base and various LBO-coated LiMn_2O_4 spinel cathode material are listed in Table 2. The amounts of Mn dissolution of base LiMn_2O_4 was obviously

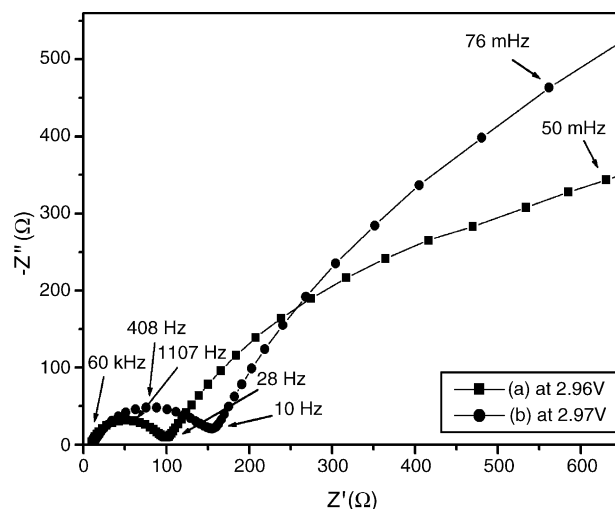


Fig. 4. Electrochemical impedance spectra of Nyquist plots for various powders calcined at 500 °C for 10 h before cycling (a) uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ and (b) 0.3 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

higher than that of the LBO-coated ones. The decrease of Mn dissolution from 128 ppm for base LiMn_2O_4 to 71.8 ppm for 0.5 wt.% LBO-coated LiMn_2O_4 was revealed. As a result, it is demonstrated that the LBO coating film could effectively reduce the dissolution of cathode electrode into the electrolyte.

Fig. 4 exhibits the Nyquist plots of electrochemical impedance spectra for the samples A1 and C1. In Fig. 4(b), the spectra of A1 displayed larger impedance with one semicircle. However, the spectra of C1 showed much smaller impedance with one semicircle, as shown in Fig. 4(a). It is suggested that the improvement of electrochemical performance is attributed to the decrease of Mn dissolution and side reaction leading to the electrolyte oxidation [34]. Furthermore, the interface area would also be reduced by surface treatment to retard the side reaction between the electrode and electrolyte, as reported in a previous work [32].

After cycling for 25 cycles, Fig. 5 reveals the Nyquist plots of electrochemical impedance spectra for samples A2 and C2. Fig. 5(a) shows that the impedance of A2 was smaller than A1 in Fig. 4(b). Nevertheless, the spectra of C2 in Fig. 5(b) showed almost the same impedance as C1 in Fig. 4(a), while smaller impedance than A2 was indicated in Fig. 5(a). It was reported that the $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass material was a good conductor in transferring the Li^+ ion [25,26]. Therefore, it is expected that the LBO coating layer with high Li^+ ionic conductor on the surface of the LiMn_2O_4 cathode material may encourage the Li^+ ion transferring through the interfaces between the LiMn_2O_4 cathode particles. This, in turn, leads to the decrease of the charge voltage, and the increase of the discharge voltage.

Table 2

ICP results of base and various LBO-coated LiMn_2O_4 powder immersed in LiPF_6 with a 1:1 (volume ratio) mixture of EC/DMC

	Base LiMn_2O_4	0.1 wt.% LBO	0.2 wt.% LBO	0.3 wt.% LBO	0.4 wt.% LBO	0.5 wt.% LBO
The amount of Mn dissolution (ppm)	128	87.1	80.8	79.1	76.4	71.8

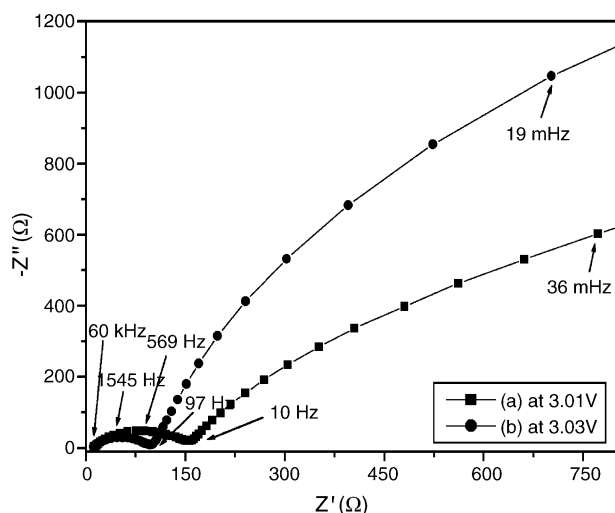


Fig. 5. Electrochemical impedance spectra of Nyquist plots for various powders calcined at 500 °C for 10h followed by 25 cycles at 25 °C: (a) uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, (b) 0.3 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

3.3. Electrochemical property at elevated temperature

The coin cell was measured galvanostatically between 3 and 4.2 V at 60 °C with the assembly of $\text{Li}/\text{EC}+\text{DMC}+\text{LiPF}_6/\text{LiMn}_2\text{O}_4$ at 0.1 C rate for the first cycle and then at 0.2 C rate after the second cycle. The fading rate at temperature as high as 60 °C was calculated from the test of base $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ and 0.1 wt.% of LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ calcined at 500 °C for 10h, as shown in Fig. 6.

Fig. 6(a) presents that the initial discharge capacity of A3 was 115.9 mAh g^{-1} at the first cycle and decayed to 84.7 mAh g^{-1} after 25 cycles. The initial discharge capacity of B2 was 117.6 and decayed to 92.44 mAh g^{-1} after 25 cycles, as shown in Fig. 6(b). It is noted that the fading rate is 16% and 13% for A2 and B2, respectively, at room temperature, i.e. there is a 3% difference in fading rate. It appears that the variation of fading

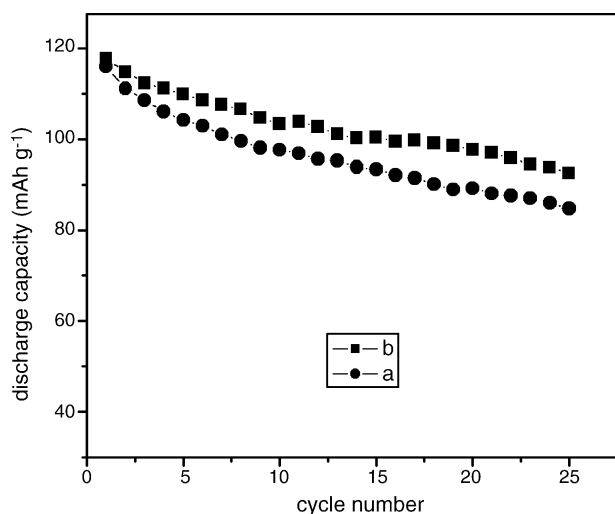


Fig. 6. Specific discharge capacity for various powders calcined at 500 °C for 10h followed by 25 cycles at 60 °C: (a) uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, (b) 0.1 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

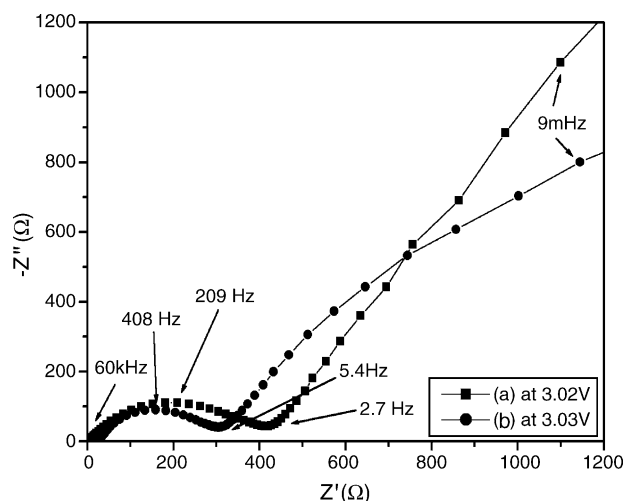


Fig. 7. Electrochemical impedance spectra of Nyquist plots for various powders calcined at 500 °C for 10h followed by 25 cycles at 60 °C: (a) uncoated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, (b) 0.1 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

rate at higher temperature of 60 °C is much appreciable since the capacity fading is 27% and 21% for A3 and B3, respectively at 60 °C, i.e. a 6% change.

Fig. 7 exhibits the Nyquist plots of electrochemical impedance spectra for the samples A3 and B3. It is apparent that the spectra of B3 show smaller impedance than A3. Moreover, in Fig. 7(a) and (b), the spectra of A3 and B3 demonstrated larger impedance with one semicircle as compared with A2 and B2. The poor electrochemical performance at high temperature is attributed to the fact that the spinel LiMn_2O_4 is particularly unstable at high temperature in Li-based electrolytes [35], resulting in dramatical Mn dissolution.

In this study, several evidences were provided to demonstrate the mechanism of the suppression of Mn dissolution and side reaction between the electrode and electrolyte by surface modification. Fig. 8 presents the X-ray diffraction pattern of 0.1 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ after 25 cycling test, indicat-

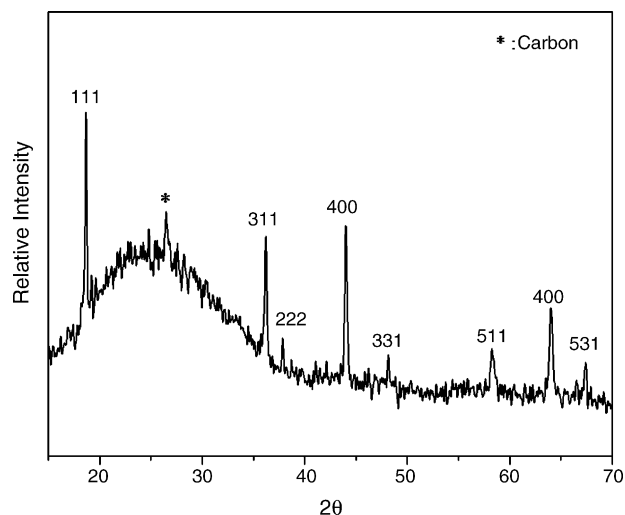


Fig. 8. X-ray diffraction pattern of 0.1 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ calcined at 500 °C for 10h followed by 25 cycles at 25 °C.

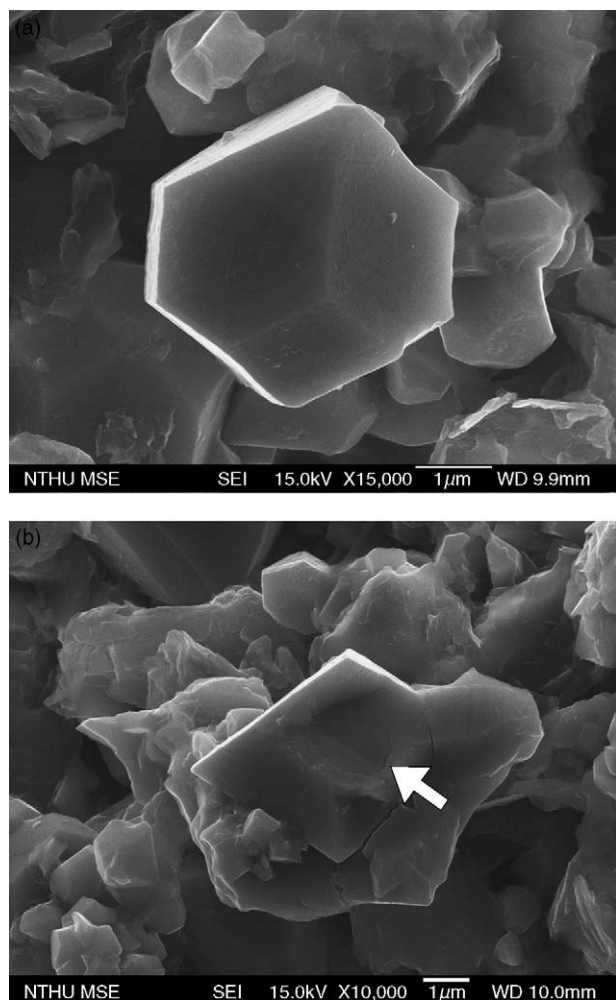


Fig. 9. FESEM images of various powders calcined at 500 °C for 10 h followed by 25 cycles at 25 °C: (a) 0.3 wt.% LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$, (b) base $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

ing spinel lithium manganese oxide phase without the formation of other phases during cycling test. Hence, the side reaction leading to the Mn dissolution and Mn_2O_3 phase [34] is thus reduced.

In fact, the capacity loss of the conventional LiMn_2O_4 cathode material is proposed to be the reaction of fluorinated anions in electrolyte with residual H_2O , which resulted in the formation of HF [24]. In fact, the amount of HF contents is a critical factor in the MnO dissolution. As shown in Fig. 9(b), the base $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ particle cycled for 25 cycles with octahedral structure was etched by HF acids. However, in Fig. 9(a), the LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ particle maintained the originally well-developed octahedral structure after 25 cycles. Hence, reducing the dissolution of Mn could be achieved by decreasing the HF contents at the interface between the cathode electrode and electrolyte.

The achievement in this study illustrates that the LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$ cathode material exhibits relatively good stability and electrochemical performance by correlating the cycling mechanism and electrochemical properties. It is believed that the improved electrochemical performance of the surface-modified

lithium manganese oxide cathode material should be partially attributed to the $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass coating.

4. Conclusions

- (1) The SEM photo and TEM image revealed that an amorphous $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass film with various weight percent was successfully coated homogeneously on the surface of $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ by solution method. The XRD pattern of the spinel LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ without other phases present also indicated that the LBO glass was only coated on the surface rather than diffusing into the crystal.
- (2) Among various amount of LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material, the 0.3 wt.% LBO-coated one exhibited the best electrochemical performance.
- (3) The side reaction and Mn dissolution between the interface of the cathode electrode and electrolyte was reduced significantly by surface modification of LBO glass in the LiMn_2O_4 .

Acknowledgments

The authors are grateful to the Coremax Taiwan Corporation, Taiwan for the financial support. Special thanks to General Manager Mr. Jim Ho for his kindness to endorse the project. Partial support from National Science Council, Taiwan, under the contract no. NSC-92-2216-E007-037 and NSC-93-2216-E007-014 are also acknowledged.

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